

TAJAMUL ASHRAF

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RESEARCH INTERESTS

Multimodal Agents, Vision-Language Reasoning, Medical AI, Video Understanding, Embodied AI, Visual Grounding, Domain Generalization, and AI for Science

EDUCATION

Johns Hopkins University

PhD in Computer Science

Advisers: **Prof. Vishal Patel** and **Prof. Daniel Khashabi**

Research focus: *Agentic Multimodal Reasoning for Cataract Surgery Videos*

Sep 2026 – Present

Baltimore, USA

Indian Institute of Technology Delhi

Master's (Research) in Information Technology, GPA: 9.28/10.0

Advisor: **Prof. Chetan Arora**

Thesis: *Domain Adaptation in Breast Cancer Detection from Mammograms.*

Aug 2022 – June 2024

New Delhi, India

National Institute of Technology Srinagar

B.Tech. in Information Technology, GPA: 8.34/10.0

Advisor: **Prof. Janibul Bashir**

Thesis: *Robust DNNs for Analyzing the Impact of Coherent Climatic Factors.*

Aug 2018 – July 2022

Srinagar, India

SELECTED PUBLICATIONS

🏆 Best Paper Award, * equal contribution, † my role as mentor

- J2** Komal Kumar*, **Tajamul Ashraf***, Omkar Thawakar, Rao Anwer, Hisham Cholakkal, Mubarak Shah, Ming-Hsuan Yang, Phillip H. S. Torr, Fahad Shahbaz Khan, Salman Khan, “*LLM Post-Training: A Deep Dive into Reasoning Large Language Models.*” In **IEEE TPAMI, 2026**. [📄 Paper](#), [🔗 Code](#)
- C9** Rozain Shakeel, Abdul Rahman Mohammad Ali, Muneeb Mushtaq, Tausifa Jan Saleem, **Tajamul Ashraf†**, “*MedSPOT: A Workflow-Aware Sequential Grounding Benchmark for Clinical GUI.*” In **ECCV’26**. [🔗 Code](#) [📄 Paper](#)
- J1** **Tajamul Ashraf**, Rajes Manna, Partha Sarathi Purkayastha, Tavaheed Tariq, Janibul Bashir, “*Context Aware Grounded Teacher for Source Free Object Detection.*” In **IJCV, 2026**. [🔗 Code](#) [📄 Paper](#)
- C8** Akash Ghosh, **Tajamul Ashraf†**, Rishu Kumar Singh, Numan Saeed, Sriparna Saha, Xiuying Chen, Salman Khan, “*CarePilot: A Multi-Agent Framework for Long-Horizon Computer Task Automation in Healthcare.*” In **CVPR’26 Findings**. [🔗 Code](#) [📄 Paper](#)
- C7** **Tajamul Ashraf***, Amal Saqib*, Hanan Gani, Muhra AlMahri, Yuhao Li, Noor Ahsan, Umair Nawaz, Jean Lahoud, Hisham Cholakkal, Mubarak Shah, Philip Torr, Fahad Shahbaz Khan, Rao Muhammad Anwer, Salman Khan, “*Agent-X: Evaluating Deep Multimodal Reasoning in Vision-Centric Agentic Tasks.*” In **ICLR’26**. [🔗 Code](#) [📄 Paper](#) [📄 Data](#)
- C6** **Tajamul Ashraf**, Janibul Bashir, “*TITAN: Query-Token based Domain Adaptive Adversarial Learning.*” In **ICCV 2025**. [🔗 Code](#) [📄 Paper](#)
- C5** Jinhong Wang*, **Tajamul Ashraf*†**, Zongyan Han, Jorma Laaksonen, Rao Anwer, “*MIRA: A Novel Framework for Fusing Modalities in Medical RAG.*” In **ACM Multimedia 2025**. [🔗 Code](#) [📄 Paper](#)
- C4** Bhuvan Sachdeva, Naren Akash, **Tajamul Ashraf**, Simon Müller, Thomas Schultz, Maximilian W. M. Wintergerst, Niharika Singri, Kaushik Murali, Mohit Jain, “*Phase-Informed Tool Segmentation for Manual Small-Incision Cataract Surgery.*” In **MICCAI 2025**. [📄 Paper](#)
- C3** **Tajamul Ashraf**, Krithika Rangarajan, Mohit Gambhir, Richa Gauba, Chetan Arora, “*D-MASTER: Mask Annealed Transformer for Unsupervised Domain Adaptation in Breast Cancer Detection from Mammograms.*” In **MICCAI 2024**. [🔗 Code](#) [📄 Paper](#)

- C2** 🏆 **Tajamul Ashraf**, Tisha Madame, “*HF-Fed: Hierarchical Based Customized Federated Learning Framework for X-Ray Imaging.*” In **MICCAI 2024** Workshop (Deep-Brea³th). [🔗 Code](#) [📄 Paper](#)
- C1** **Tajamul Ashraf**, Fuzayil Bin Afzal Mir, Iqra Altaf Gillani, “*TransFed: A Way to Epitomize Focal Modulation Using Transformer-Based Federated Learning.*” In **WACV 2024**. [🔗 Code](#) [📄 Paper](#)

RESEARCH EXPERIENCE

Johns Hopkins University

Sep 2026 – Present

Research Assistant Hosts: Prof. Daniel Khashabi and Prof. Vishal Patel

Baltimore, USA

- Developing video-language agents that reason over temporal surgical events, track procedural state, and produce grounded, verifiable feedback.

Nanyang Technological University (NTU)

Mar 2026 – Aug 2026

Research Manager Host: Prof. Adams Wai-Kin Kong

Singapore

- Collaborated with the **INTERPOL** team on video understanding and reasoning for surveillance.
- Developed multimodal agentic systems for long-horizon temporal reasoning and decision-making.

King Abdullah University of Science and Technology (KAUST)

Dec 2025 – Mar 2026

Research Consultant Host: Prof. Bernard Ghanem

Remote

- Introduced **MedCTA**, a benchmark for evaluating multimodal clinical tool agents.
- Developed evaluation protocols for grounded and reliable tool-augmented clinical reasoning.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

Jan 2025 – Nov 2025

Research Engineer Hosts: Prof. Salman Khan, Prof. Rao Muhammad Anwer

Abu Dhabi, UAE

- Curated **Agent-X**, a large-scale benchmark for perception, grounding, and decision-making in real-world agentic tasks; the project won **2nd place in the Reasoning & Planning Track** of the Berkeley AgentX Competition 2025 as Team Agent-VBench.
- Built a multimodal RAG framework (**MIRA**) for factual and adaptive medical agents.
- Authored a survey on post-training techniques for reasoning LLMs.

Microsoft Research (MSR)

July 2024 – Dec 2024

Research Intern Hosts: Dr. Mohit Jain, Dr. Prachi Jain

Bangalore, India

- Built an LLM-based knowledge-graph and RAG framework over a semi-structured knowledge base for medical chatbots.
- Jointly introduced the *Sankara-MSICS* dataset and proposed a phase-conditioned method with pseudo-labeling for surgical tool segmentation.

AI & Robotics Technology Park (ARTPARK), I-Hub

Dec 2021 – May 2022

Visiting Student Host: Prof. Raghu Krishnapuram

Bangalore, India

- Developed an automated washroom-cleaning robot, *RoboJanitor*; implemented pose estimation, detection, classification, and SLAM.
- Built 3D pose models and REST APIs for production deployment.

Indian Institute of Science (IISc)

Mar 2021 – Sep 2021

Research Intern Host: Prof. S. N. Omkar

Bangalore, India

- Built a human-pose comparison model to assist yoga instruction in live online settings.
- Used BlazePose for keypoint extraction and delivered a feedback mechanism for real-time pose correction.

Teaching Assistant

Probability and Statistics (SC2000), NTU Singapore (Spring 2026), instructed by **Prof. Adams Kong**.

Trustworthy Medical Vision (CV8502), MBZUAI (Fall 2025), instructed by **Prof. Yutong Xie**.

Computer Vision for Robotics (JRL780), IIT Delhi (Spring 2024), instructed by **Prof. Chetan Arora**.

Computer Science Fundamentals and Programming (COL100), IIT Delhi (Fall 2023), instructed by **Prof. Chetan Arora** and **Prof. Vireshwar Kumar**.

Data Structures and Algorithms (COL106), IIT Delhi (Fall 2022), instructed by **Prof. Naveen Garg** and **Prof. Ashish Chiplunkar**.

Reviewer

2027	ICLR, IJCV
2026	NeurIPS, ACM Multimedia, MICCAI, IJCV
2025	ICCV, ACM Multimedia, MICCAI, WACV, Pattern Recognition, IJCV, TAI
2024	MICCAI, WACV, CVIP, TPDS
2023	CVIP, HPEC

Professional Service and Volunteering

Program Committee Member: ACM Multimedia 2026 (Rio de Janeiro, Brazil).

Volunteer: ICLR 2026 (Rio de Janeiro, Brazil), ACM COMPASS 2025 (Toronto, Canada), CVIP 2023 (IIT Jammu, India).

Invited Talks

Invited speaker at *Sangam 2.0* (NIT Srinagar Alumni Meet), delivering a talk on *Research Directions in the Current AI Landscape*, covering research opportunities, higher studies, and emerging trends in AI (2025).

Delivered a keynote talk on “Multimodal Systems for Vision-Language Domains in Advancing Healthcare” at the BRAIN-AID Seminar, University of Lagos, Nigeria (April 2025).

Gave two talks on undergraduate research directions and campus placements at NIT Srinagar (June 2024).

Mentorship

To give back, I co-founded Gaash Research Lab to foster a research culture among students and actively mentor underrepresented students on thesis projects and publication-oriented research. Non-exhaustive list:

2026

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| 14. Ilakkiya Selvakumar (NTU Singapore) | <i>Benchmarking Multimodal Agents for Surveillance Applications</i> |
| 13. Uzair Nazir, Abdul Muizz (NIT Srinagar) | <i>OCR and Kashmiri Language Translation</i> |
| 12. Adab Ismail (NIT Srinagar) | <i>Vision-Language Modeling for Object Tracking</i> |
| 11. Sheikh Samee ul Haq, Aabid Nabi (NIT Srinagar) | <i>Robust Tool Segmentation in Surgical Scenarios</i> |
| 10. Rozain Shakeel, Muneeb Ahmad (NIT Srinagar) | <i>Grounding and Reasoning for Clinical GUI</i> |
| 9. Tafheem Malik, Iram Naseer (NIT Srinagar) | <i>Multimodal Image and Video Generation Using Reasoning</i> |
| 8. Ifrah Mushtaq (University of Kashmir) | <i>ASR and Phonetic Analysis for Kashmiri Language</i> |

2025

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| 7. Burhaan Rasheed Zargar (NIT Srinagar) | <i>Script-Aware Flow Matching for Kashmiri Text-to-Speech</i> |
| 5. Abrar Riyaz (NIT Srinagar), Wasif Tak (Thapar University) | <i>Language-Conditioned Surgical Tool Segmentation</i> |
| 4. Tawaheed Tariq, Sonia Yadav (NIT Srinagar) | <i>Query-Driven Reasoning for Multimodal MOT</i> |

2024

3. Suhaib Salmani (NIT Srinagar) *VLM-Based Masked Learning for Efficient Medical Object Detection*
2. Rajes Manna (NIT Srinagar) *Context Aware Grounded Teacher for Source-Free Object Detection*

2023

1. Asrar ul Haq (NIT Srinagar) *Transformer-Based Unsupervised Domain Adaptation in a Federated Setup*

ACHIEVEMENTS AND HONORS

- Awarded the **Johns Hopkins PhD Fellowship** and **Summer Relocation Grant** (Baltimore, USA).
- **ICLR 2026 Registration and Travel Grant** (Rio de Janeiro, Brazil).
- **MICCAI 2025 Outstanding Reviewer Award**.
- Awarded travel and registration grants for **ICCV 2025** (Honolulu, Hawaii), **ACM Multimedia 2025** (Dublin, Ireland), and the **Agentic AI Summit 2025** (UC Berkeley, California).
- **Second Place, Reasoning & Planning Track**, Berkeley AgentX Competition 2025, presented at the Agentic AI Summit, UC Berkeley (Team Agent-VBench).
- Secured 2nd place in the MBZUAI Department Logo Design Competition.
- Awarded a \$500 compute grant from **alphaXiv** for contributions to LLM research.
- **Best Paper Award**, Deep-Brea³th Workshop at MICCAI 2024 (HF-Fed).
- Nominated as a **Diversity Intern** by Microsoft Canada.
- Awarded the **IIT Delhi Endowment Grant** and a travel grant for **MICCAI 2024** (Marrakesh, Morocco).
- **Microsoft Learn Student Ambassador – Gold**, NIT Srinagar.
- Won the **Advanced Track of the National Entrepreneurship Challenge** at IIT Bombay (2021; INR 90,000 team cash prize).

MEDIA APPEARANCES

- *NIT Srinagar Hosts KATHE 2026 Challenge to Develop English-to-Kashmiri Translation.* [🔗](#)
- *A New Stress Test for AI Agents That Plan, Look, and Click.* [🔗](#)
- *Meet a Microsoft Learn Student Ambassador Graduate: Tajamul Ashraf.* [🔗](#)
- *NIT Srinagar Students Win India's Largest National Entrepreneurship Challenge.* [🔗](#)
- *Hult Prize: A Futuristic Approach to Entrepreneurship.* [🔗](#)
- *Workshop on Entrepreneurship, Research and Innovation at NIT Srinagar.* [🔗](#)

SKILLS & INTERESTS

Technical Skills	Python, C/C++, PyTorch, DeepSpeed, FSDP, Model/Sequence Parallelism
Leadership / Outreach	Founder of Ralith Milth (anti-drug NGO), Co-founder Gaash Lab (Mentoring)
Personal Interests	Badminton, swimming, blog (Journey with Bytes)

REFERENCES

[Prof. Adams Kong](#) (NTU Singapore) [Prof. Daniel Khashabi](#) (Johns Hopkins) [Prof. Vishal Patel](#) (Johns Hopkins)